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PART B – TRAFFIC CONTROL

Section 11: Traffic Control Layouts – Intersections

LEGEND			
	Flexible Drum		Paint Truck
	Tubular Marker		Escort Truck
	Cone		Chaser Vehicle
	Sign		Vehicle Mounted Rear Crash Attenuator
	Traffic Control Person		
	Work Activity Area		360° Flashing Light
	Work Truck		Portable Traffic Signal
	Shadow Vehicle		Barricade and Fencing
	Shadow Vehicle #1		Flashing Arrow Board (FAB)
	Shadow Vehicle #2		Flashing Arrow Board (FAB) in caution mode
	Buffer Vehicle		Dynamic Message Sign (DMS)

TABLE A – TAPER LENGTHS

Taper Types (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Merge Taper Length	L_M	35	55	160	190	210	230	250	280
Lane Shift Taper Length	L_L	30	50	80	100	110	120	130	140
Downstream Taper Length	L_D	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	L_S	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	L_T	30	60	160	190	210	230	250	280
Run-In Length on Centreline	L_R	40	50	60	60	70	80	90	100

TABLE B – DEVICE SPACING LENGTHS

Device Spacing (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Construction Sign Spacing	A	40	60	80	100	150	150	200	200
Buffer Space	B	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	R	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	C	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	D	10	10	30	30	40	40	40	50

11.1 General Information - Intersections[1]

1. Traffic control within an intersection requires careful consideration of all the approaches and accesses, the nature of adjacent land uses, and the intersection control type.
2. To control each leg of an intersection, multiple Traffic Control Persons are typically required.
3. Signalized intersections typically exist on high-volume corridors that may experience peak traffic periods. This shall be considered when determining the appropriate traffic control for the project. If excessive queues develop, traffic control should be removed as soon as possible to restore normal traffic operations.
4. Within a signalized intersection, the direction provided by Traffic Control Persons cannot conflict with the direction provided by the signal heads so the signal shall be either turned off or bagged.
5. Good communication between the Traffic Control Persons who are operating an intersection helps to maintain reasonable delay periods on each approach and ensures that multiple traffic streams are not simultaneously given conflicting right-of-way through the intersection.
6. Careful consideration should also be given to other road users, such as pedestrians and cyclists, to ensure that they can negotiate through the intersection satisfactorily. This may require the use of additional Traffic Control Persons who are dedicated to serving pedestrian demand.

[1] See *Section 12: Traffic Control Layouts – Roundabouts* for information on traffic control layouts for roundabouts.

The traffic control layouts are considered the minimum standard. The associated text description of each figure highlights the key standards as well as guidance and options that can be considered by the user. The following information is typical on most layouts in this section:

Standard:

- For short-duration work, a Crew Working Ahead C-004 sign shall be used.
- For long-duration work, a Construction Ahead C-018-1A sign shall be used.
- Traffic Control Person Ahead C-001-1 signs shall be used in advance of TCPs.
- On undivided roadways, barricades are required at each end of the work activity area for long-duration work. They are required on the approach side only for divided roadways.
- Where the speed limit is ≥ 70 km/h, a buffer space shall be used.
- All work, buffer, and shadow vehicles shall be equipped with a 360-degree flashing light and 4-way flashers.

Guidance:

- Where the speed limit is ≥ 70 km/h a buffer vehicle should be used when workers are present.
- A vehicle-mounted crash attenuator is recommended for use on buffer vehicles.
- When a detour is not implemented, to maintain traffic flow, it may be necessary to restrict turning movements at the intersection (i.e., prohibit left and/or right turns using Turn Control R-015-L/R signs).

Options:

- The Flagger Ahead C-001-2 sign or Prepare to Stop C-029 sign may be used for additional advance warning where TCPs are stopping traffic
- A portable dynamic message sign (DMS) may be used to provide advance messaging for drivers.
- When traffic volumes are high or the intersection is signalized, consult the Road Authority.
- In low speed (≤ 60 km/h) urban areas (within municipal boundaries), customization of traffic control layouts, including closer device spacing and shorter taper lengths, may be necessary to maintain access and mobility. In these cases, document why adjustments are being made.

11.2 Intersection Lane Closure – Two-Lane, Two-Way Roadway with TCPs (Near Side) – Short and Long Duration

Purpose:

This layout shows the typical setup for a lane closure using Traffic Control Persons on a two-lane, two-way roadway when the closure is on the approach to (near side of) an intersection. It may be used where the intersection is signalized or stop-controlled.

One TCP is typically needed for each leg of the intersection because traffic control is complex at intersections, especially for turning traffic.

Standard:

- Single Lane Traffic C-030-8 signs are required in both directions along the roadway where the work is being conducted.
- A Prepare to Stop C-029 sign shall be placed on the cross street in advance of the intersection.
- When TCPs are directing traffic, the construction speed limit shall be ≤ 70 km/h.

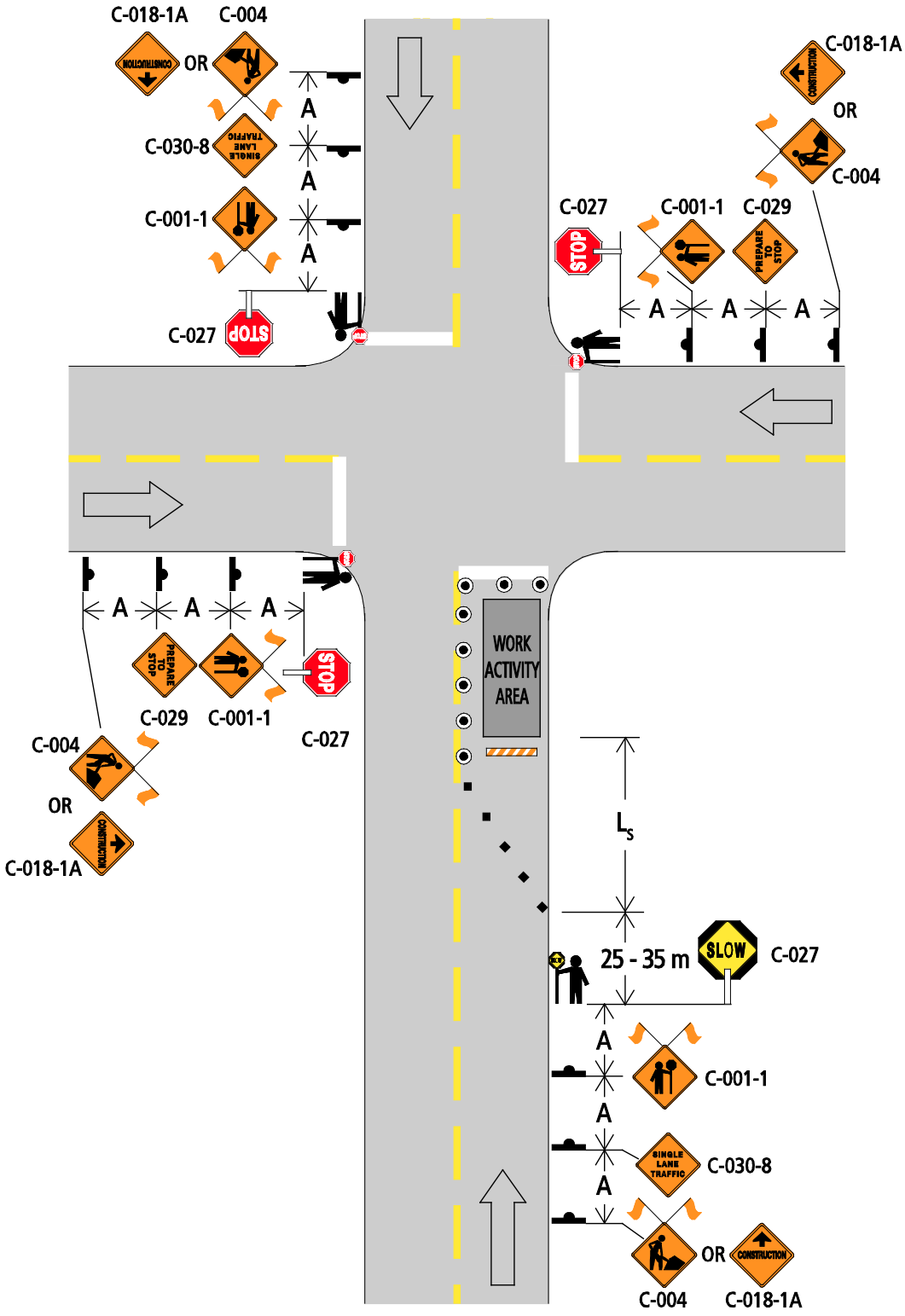
Guidance:

- None at this time.

Options:

- Where approach speeds are ≤ 60 km/h, cones may be used instead of tubular markers.
- If the-cross street volume is low, TCPs may not be required in the cross direction but Traffic Control Person Ahead C-001-1 signs are still required to identify the presence of other TCPs controlling traffic.

Figure 11.2: Intersection Lane Closure – Two-Lane, Two-Way Roadway with TCPs (Near Side) – Short and Long Duration



11.3 Intersection Lane Closure – Two-Lane, Two-Way Roadway with Detour (Near Side) – Short and Long Duration

Purpose:

This layout shows the typical setup for a lane closure on a two-lane, two-way roadway when the closure occurs on the approach to (near side of) an intersection. It may be used where a detour can be established to allow drivers to bypass the closed area. Detour signage is continued at decision points through the detour route to guide motorists along the detour and back to the main roadway.

This layout is to be used only if a detour route is available.

If no alternative route is available, TCPs are required as shown in *Section 11.2: Intersection Lane Closure – Two-Lane, Two-Way Roadway with TCPs (Near Side).*

Standard:

- Either a Road Closed Ahead C-030-6A sign or a dynamic message sign (DMS) shall be the first sign used on the approach to the closed portion of roadway.
- A Detour Ahead Left/Right C-006-L/R marker shall be used in advance of the detour route to guide traffic in the appropriate direction.
- A Turn Control R-015-L/R sign shall be used on the cross street to restrict traffic from turning into the work area.
- Barricades on the closed lane shall display to approaching traffic a Road Closed R-012 sign and a Detour C-005-L/R sign.

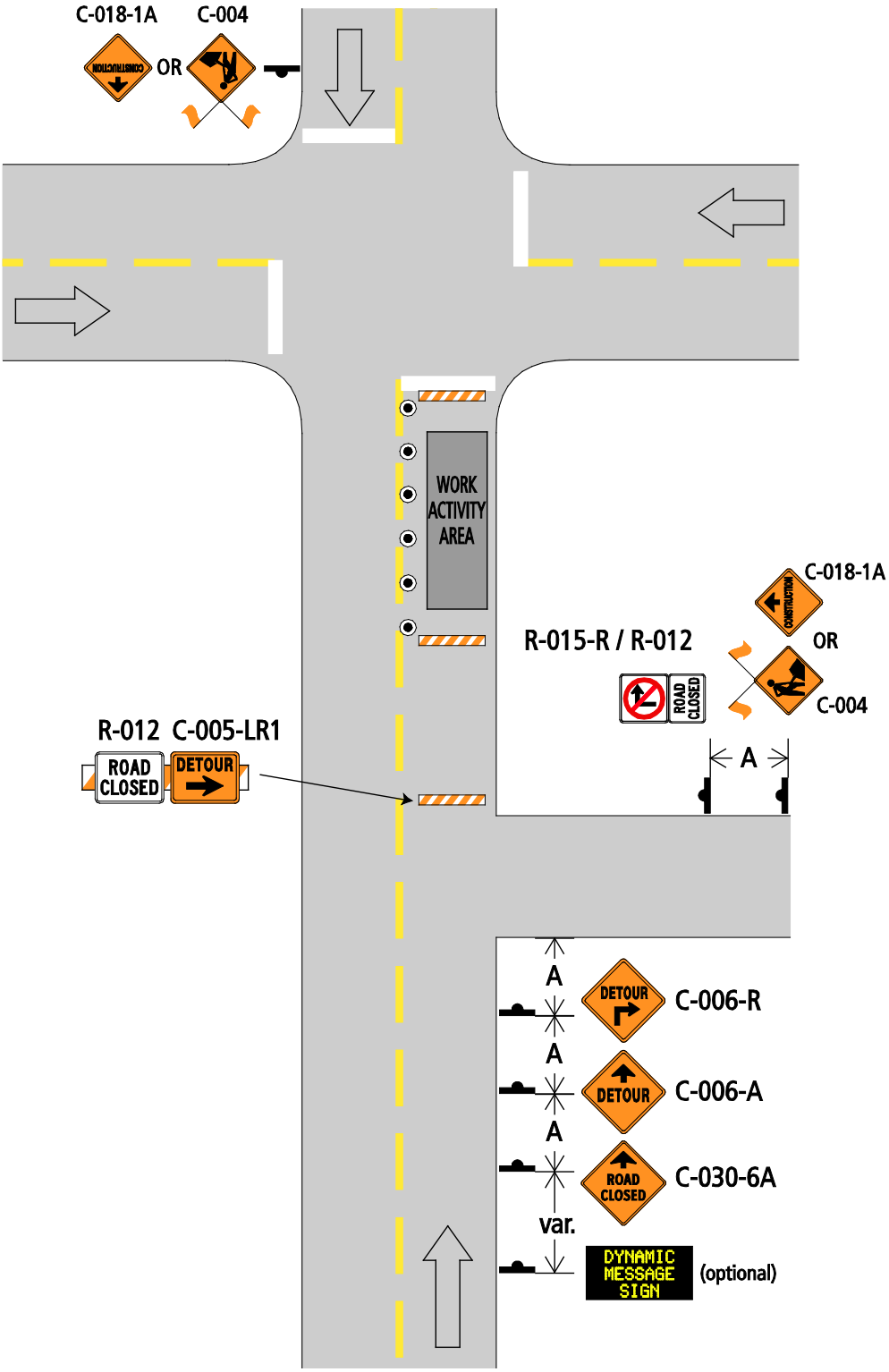
Guidance:

- Advance planning is recommended for situations involving business access and egress.
- The detour route chosen should have the least impact on the travelling public and local residents. This may necessitate establishing the detour on a road further in advance of the work area.

Options:

- Flashing arrow boards (FABs) may be used in addition to barricades. They should be set to caution mode to mark the closed portion of the roadway, and to arrow mode at the detour.

Figure 11.3: Intersection Lane Closure – Two-Lane, Two-Way Roadway with Detour (Near Side) – Short and Long Duration



11.4 Intersection Lane Closure – Two-Lane, Two-Way Roadway with TCP's (Far Side) – Short and Long Duration

Purpose:

This layout shows the typical setup for a lane closure using Traffic Control Persons on a two-lane, two-way roadway when the closure occurs downstream (far side) of an intersection. It may be used where the intersection is signalized or stop-controlled.

One TCP is typically needed for each leg of the intersection because traffic control is complex at intersections, especially for turning traffic.

Standard:

- Single Lane Traffic C-030-8 signs are required in both directions along the roadway where the work is being conducted.
- A Prepare to Stop C-029 sign shall be placed on the cross street in advance of the intersection.
- A flashing arrow board (FAB) in caution mode shall be placed inside the approaching taper in advance of the work area.
- Tubular markers are required for channelizing traffic and for tapers.
- When TCPs are directing traffic, the construction speed limit shall not be greater than 70 km/h.

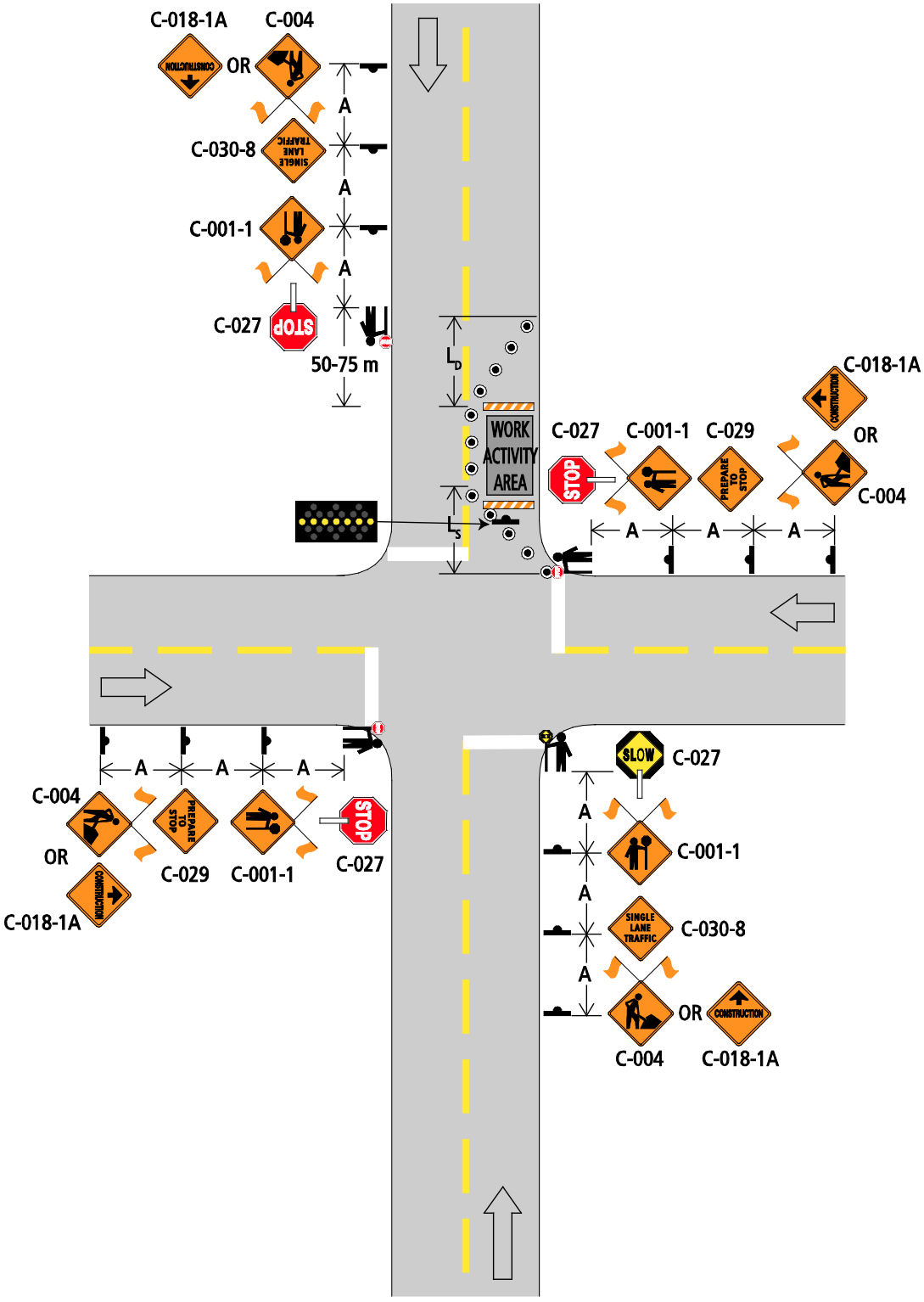
Guidance:

- None at this time.

Options:

- Where approach speeds are ≤ 60 km/h, cones may be used instead of tubular markers.
- If the speed of the roadway on which the work is being undertaken is ≤60 km/h, the flashing arrow board (FAB) may be replaced by a barricade with a flashing light.
- If the cross-street volume is low, TCPs may not be required in the cross direction but Traffic Control Person Ahead C-001-1 signs are still required to identify the presence of other TCPs directing traffic.

Figure 11.4: Intersection Lane Closure – Two-Lane, Two-Way Roadway with TCPs (Far Side) – Short and Long Duration



11.5 Intersection Lane Closure – Two-Lane, Two-Way Roadway with Detour (Far Side) – Short and Long Duration

Purpose:

This layout shows the typical setup for a lane closure on a two-lane, two-way roadway when the closure occurs downstream (far side) of an intersection. It may be used where a detour can be established to allow drivers to bypass the closed area. Detour signage is continued at decision points through the detour route to guide motorists along the detour and back to the main roadway.

This layout is to be used only if a detour route is available.

If no alternative route is available, TCPs are required as shown in *Section 11.4: Intersection Lane Closure – Two-Lane, Two-Way Roadway with TCPs (Far Side).*

Standard:

- Either a Road Closed Ahead C-030-6A sign or a dynamic message sign (DMS) shall be the first sign used on the approach to the closed portion of roadway.
- A Detour Ahead Left/Right C-006-L/R sign shall be used in advance of the detour route to guide traffic in the appropriate direction.
- A Turn Control R-015-L/R sign shall be used on the cross street to restrict traffic from turning into the work area.
- Tubular markers are required for channelizing traffic and for tapers.
- A flashing arrow board (FAB) in caution mode shall be used in advance of the work area.

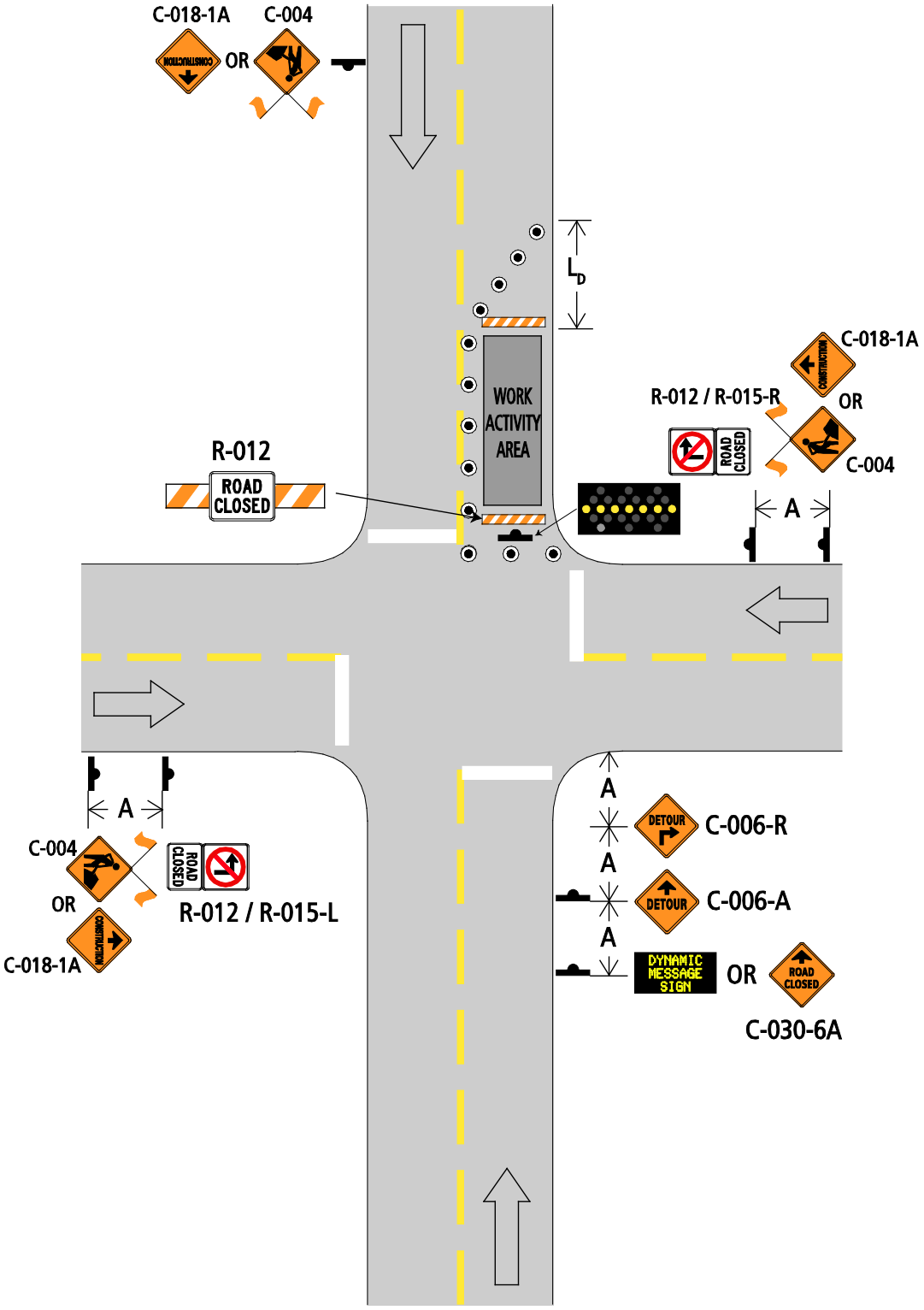
Guidance:

- Advance planning is recommended for situations involving business access and egress.
- The detour route chosen should have the least impact on the travelling public and local residents. This may necessitate establishing the detour on a road further in advance of the work area.

Options:

- Where approach speeds are ≤60 km/h, cones may be used instead of tubular markers.
- If the speed of the roadway on which the work is being undertaken is ≤60 km/h, the flashing arrow board (FAB) on the approach side of the work activity area may be replaced by a barricade with a Road Closed R-012 sign.

Figure 11.5: Intersection Lane Closure – Two-Lane, Two-Way Roadway with Detour (Far Side) – Short and Long Duration



11.6 Right Lane Closure (Near Side) – Multilane Intersection – Short and Long Duration

Purpose:

This layout shows the typical setup for a single lane closure on a multilane roadway when the closure occurs on the approach to (near side of) an intersection. It may be used where the intersection is signalized or stop-controlled.

Since at least one lane is available approaching the intersection in the affected direction, traffic is diverted into an adjacent lane through a lane drop.

Standard:

- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab is required in advance of a second Right Lane Closed Ahead C-130-R sign.
- A flashing arrow board (FAB) in arrow mode shall be placed inside the approaching taper in advance of the work area.

Guidance:

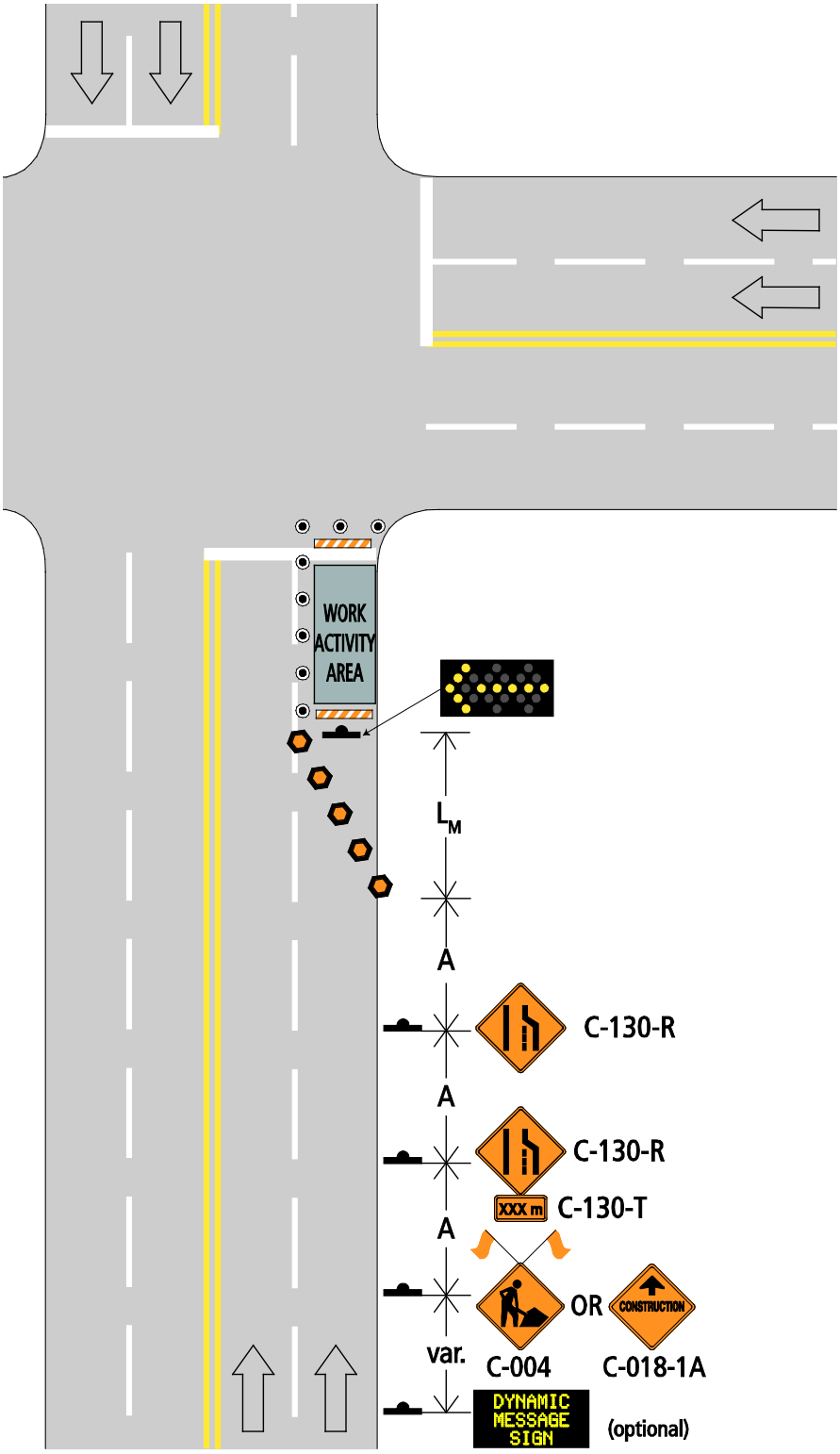
- Median-mounted signs matching the shoulder-mounted signs should be used where space allows.

Options:

- A dynamic message sign (DMS) may be used in advance of the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign.
- If the speed limit is ≤ 60 km/h:
  - The upstream Right Lane Closed C-130-R sign may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
  - Cones may be used for protecting the work area.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 11.6: Right Lane Closure (Near Side) – Multilane Intersection – Short and Long Duration



11.7 Left Lane Closure (Far Side) – Multilane Intersection – Short and Long Duration

Purpose:

This layout shows the typical setup for a single lane closure on a multilane roadway when the closure occurs downstream (far side) of an intersection. It may be used where the intersection is signalized or stop-controlled.

Since at least one lane is available approaching the intersection in the affected direction, traffic is diverted into an adjacent lane through a lane drop. Although the work is taking place downstream of the intersection, the lane drop should be established in advance of the intersection.

Standard:

- The left lane shall be closed on the near side of the intersection and traffic moved to the right lane.
- A Left Lane Closed Ahead C-130-L sign with a C-130-T distance tab is required in advance of a second Left Lane Closed Ahead C-130-L sign.
- A flashing arrow board (FAB) in arrow mode shall be placed inside the taper in advance of the work area on the near side of the intersection.
- A Lane Closure Arrow C-053 sign and barricade shall be positioned immediately in advance of the work area on the far side of the intersection.

Guidance:

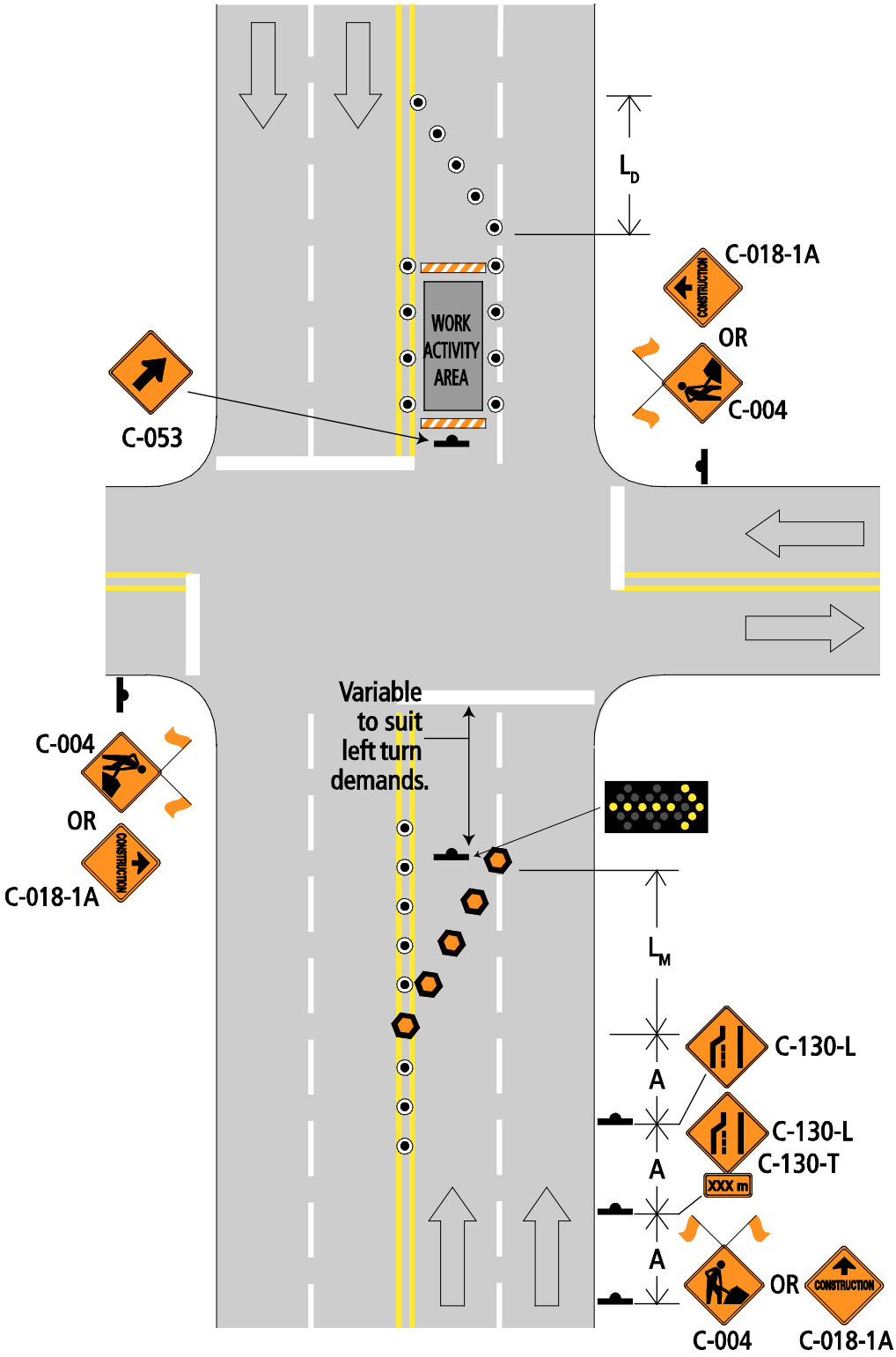
- A left lane that has significant left-turning movements may remain open as a turn lane for left turns only. Sufficient space should be considered for vehicle storage in this lane, and additional traffic control should be considered.
- Median-mounted signs matching the shoulder-mounted signs should be used where space allows.

Options:

- If the speed limit is ≤ 60 km/h:
  - The upstream Left Lane Closed Ahead C-130-L sign may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 11.7: Left Lane Closure (Far Side) – Multilane Intersection – Short and Long Duration



11.8 Right Lane Closure (Far Side) – Multilane Intersection – Short and Long Duration

Purpose:

This layout shows the typical setup for a single lane closure on a multilane roadway when the closure occurs downstream (far side) of an intersection. It may be used where the intersection is signalized or stop-controlled.

Since at least one lane is available approaching the intersection in the affected direction, traffic is diverted into an adjacent lane through a lane drop. Although the work is taking place downstream of the intersection, the lane drop should be established in advance of the intersection.

Standard:

- The right lane shall be closed on the near side of the intersection and traffic moved to the left lane.
- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab is required in advance of a second Right Lane Closed Ahead C-130-R sign.
- A flashing arrow board (FAB) in arrow mode shall be placed inside the taper in advance of the work area on the near side of the intersection.
- A Lane Closure Arrow C-053 sign and barricade shall be used immediately in advance of the work area on the far side.

Guidance:

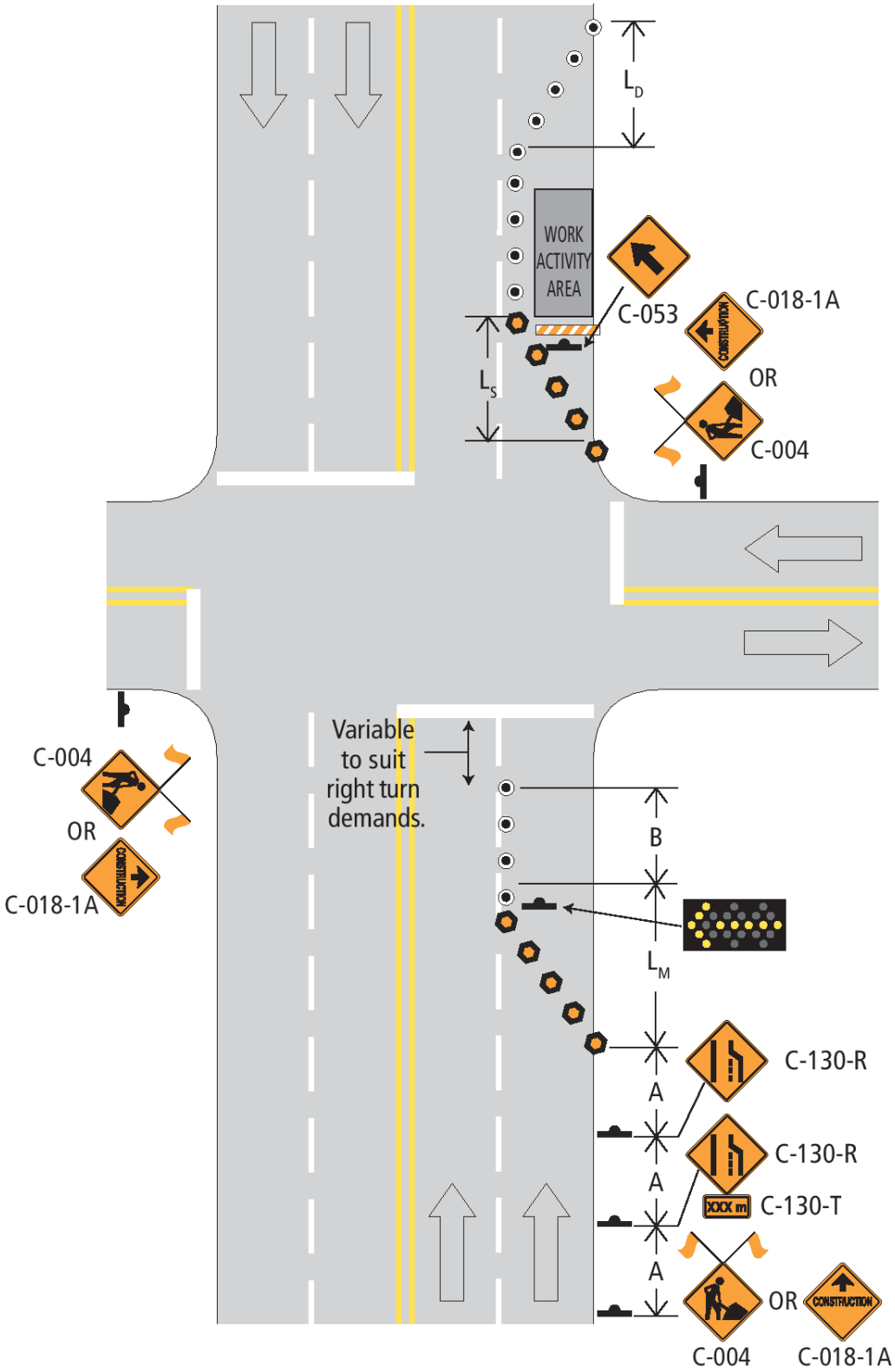
- A right lane that has significant right-turning movements may remain open as a turn lane for right turns only. Sufficient space should be considered for vehicle storage in this lane, and additional traffic control should be considered.
- Median-mounted signs matching the shoulder-mounted signs should be used where space allows.

Options:

- If the speed limit is ≤ 60 km/h:
  - The upstream Right Lane Closed Ahead C-130-R sign may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 11.8: Right Lane Closure (Far Side) – Multilane Intersection – Short and Long Duration



11.9 Left/Right Lane Closure within Intersection – Multilane Intersection – Short and Long Duration

Purpose:

This layout shows the typical setup for a single lane closure on a multilane roadway where the closure extends into—and possibly through—an intersection that affects both major road and cross-direction traffic. It may be used where the intersection is signalized or stop-controlled.

Traffic from the closed lane on the major road is diverted into an adjacent lane using a lane drop. Traffic on the cross street should be informed of the partial or complete closure, and re-routed using a detour route if possible.

Because of the lane closure within the intersection, it is necessary to restrict turning movements at the intersection for certain directions of travel (i.e., prohibit left and/or right turns by using Turn Control R-015-L/R signs).

Standard:

- On the **major road**:
  - The left/right lane shall be closed on the near side of the intersection, and traffic moved to the left/right lane.
  - A Left/Right Lane Closed Ahead C-130-L/R sign with a C-130-T distance tab is required in advance of a second Left/Right Lane Closed Ahead C-130-L/R sign.
  - A Road Closed R-012 sign and a Turn Control R-015-L/R sign shall be placed in advance of the work area and repeated closer to the intersection.
  - In the opposing direction on the major road, a Road Closed R-012 sign and a Turn Control R-015-L/R sign shall be used.
  - A flashing arrow board (FAB) in arrow mode shall be placed inside the taper in advance of the work area on the near side of the intersection.
- In the **cross-street direction**:
  - From both directions, a Road Closed Ahead C-030-6A sign or a dynamic message sign (DMS) shall be the first sign used on the approach to the closed portion of roadway.
  - Turning shall be restricted in the cross direction with a No Through Traffic R-017-2 sign and a Road Closed R-012 sign.
  - The closed lane shall be blocked by a barricade with a Road Closed R-012 sign and a Detour C-005-LR1 marker.
  - The barricade on the intersection side shall display a Road Closed R-012 sign.
- If a **detour route** is available, additional signage is required:
  - A Detour Ahead C-006-A marker shall be used in advance of the detour.
  - A Detour Ahead Left/Right C-006-LR marker shall be positioned just in advance of the detour route to guide traffic in the appropriate direction.

Guidance:

- Detour signing should be provided on both cross-street approaches wherever possible, and will vary with the detour routes available.

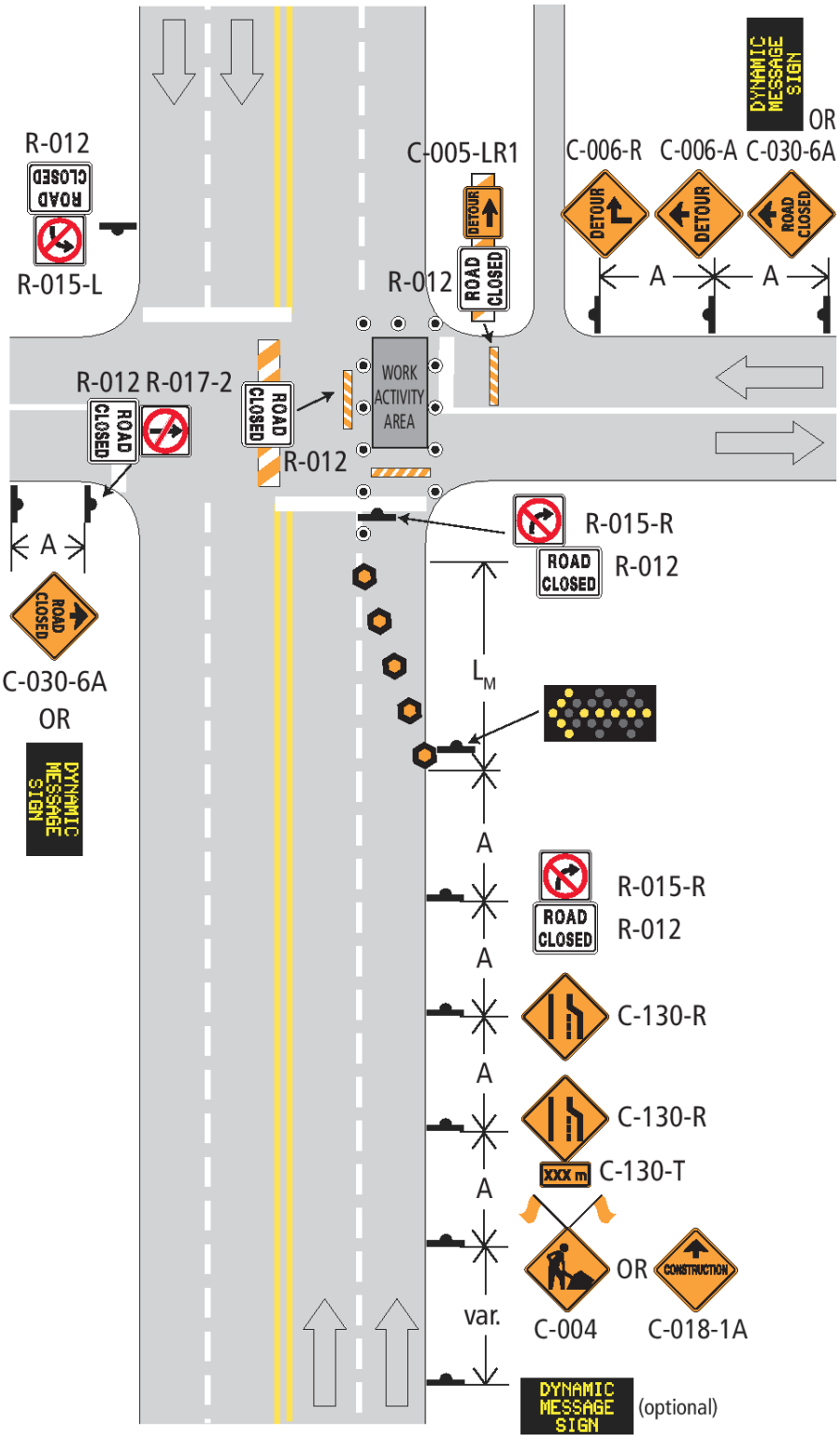
Options:

- If the speed limit is ≤ 60 km/h:
  - The upstream Left/Right Lane Closed Ahead C-130-L/R sign may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

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Figure 11.9: Left/Right Lane Closure within Intersection – Multilane Intersection – Short and Long Duration



11.10 Right Lane Closure with Right-Turn Lane (Near Side) – Channelized Right Turn Open – Multilane Intersection – Short and Long Duration

Purpose:

This layout shows the typical setup for a single lane closure on a multilane roadway when the closure occurs on the approach to (near side of) an intersection.

It is typically be used at a signalized intersection where there is a channelized right-turn lane.

Traffic in the right lane is directed into the right-turn lane. Drivers in the right lane are informed in advance that traffic in the right lane must turn right.

Standard:

- A Right Lane Must Turn Right R-082-R2 sign with a C-130-T distance tab is required in advance of a second Right Lane Must Turn Right R-082-R2 sign.
- The flashing arrow board (FAB) shall be set to caution mode.

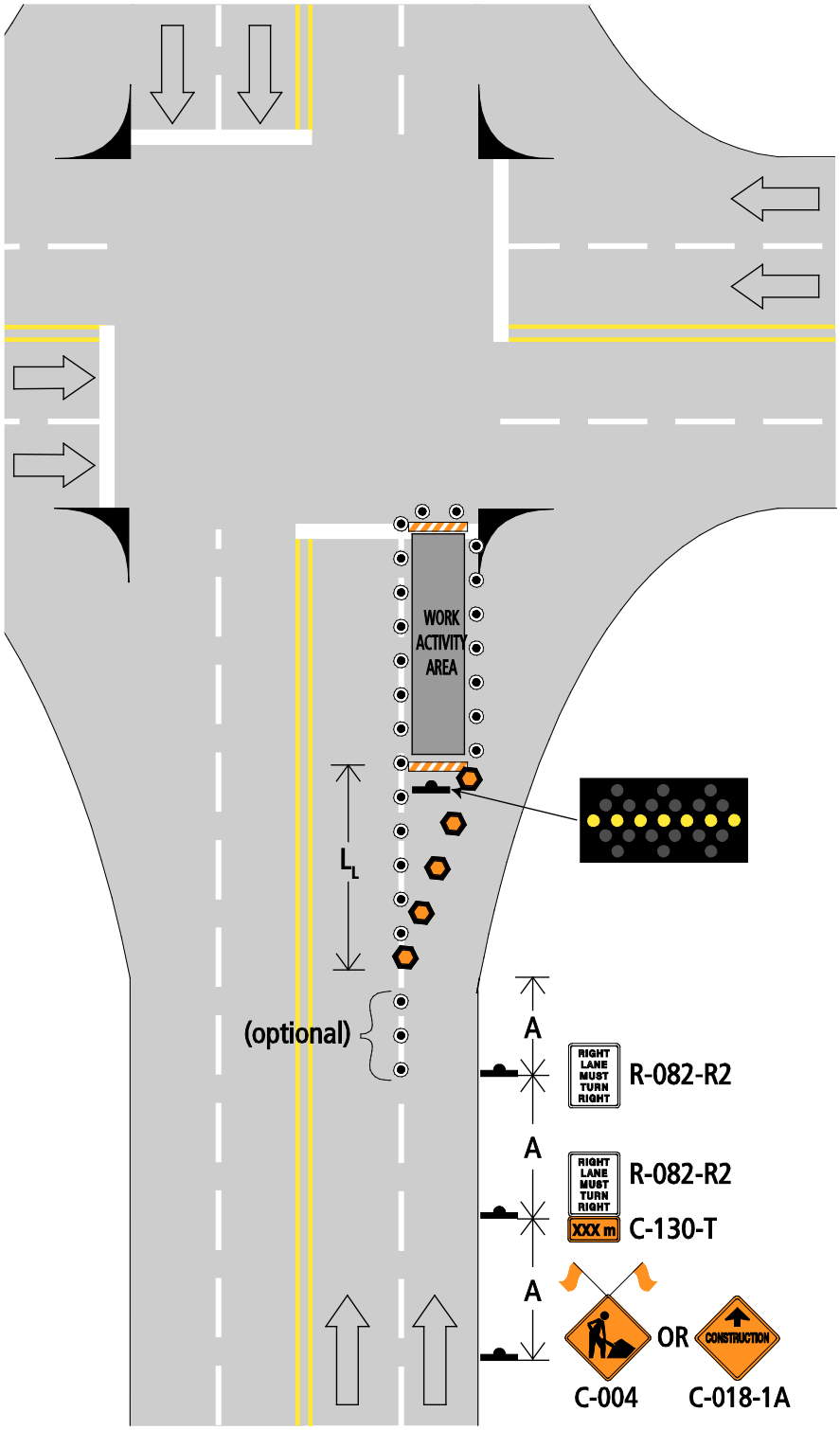
Guidance:

- To direct right-turn vehicular traffic into the right-turn lane, tubular markers or drums should be placed in a line parallel to the pavement edge to create a taper.
- Median-mounted signs matching the shoulder-mounted signs should be used where space allows.

Options:

- To separate traffic earlier in advance of the work area, additional channelizing devices may be placed along the dashed lane line to separate the left and right lanes.

Figure 11.10: Right Lane Closure with Right-Turn Lane (Near Side) – Channelized Right Turn Open – Multilane Intersection – Short and Long Duration



11.11 Two Lanes Closed (Near Side) – Multilane Intersection – Short and Long Duration

Purpose:

This layout shows the typical setup for a double lane closure on a multilane roadway when the closure occurs on the approach to (near side of) an intersection.

Both through lanes are closed, and through traffic is diverted into the left-turn lane to accommodate its movement through the intersection.

Standard:

- Traffic shall be merged into a single lane before approaching the left-turn and right-turn lanes.
- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab is required in advance of a second Right Lane Closed Ahead C-130-R sign.
- A double-sided taper shall be indicated with drums, and a Double Hazard C-154-D marker shall be positioned in advance of the work area to direct traffic to the left and right lanes.
- Flashing arrow boards (FABs) shall be positioned as follows:
  - FAB #1 in arrow mode: in the right lane drop taper in advance of the work area
  - FAB #2 in caution mode: inside the work area taper in advance of the work

Guidance:

- It is intended that traffic moving through the intersection use the left-turn lane. Signal timings may have to be adjusted in all directions at signalized intersections.
- A lane use sign, such as the Lane Use R-083-L sign, should be positioned before the intersection to provide clarity for drivers.
- Median-mounted signs matching the shoulder-mounted signs should be used where space allows.

Options:

- Temporary dashed pavement marking may be added across the intersection to guide traffic through the intersection into the downstream lane.
- If the speed limit is ≤ 60 km/h:
  - The upstream Right Lane Closed Ahead C-130-R sign may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
  - A Lane Closure Arrow C-053 sign may replace FAB #1